

DATA ENGINEERING COURSE

Library Management System

ASSIGNMENT

Instructor: Tanishq Tyagi

Student Instructions

- Read the assignment requirements carefully before writing any code.
- Your solution must be implemented using Object-Oriented Programming (OOP) principles as taught in class.
- All classes, methods, and variables must have meaningful, descriptive names.
- Follow PEP 8 formatting guidelines for all Python code.
- Your solution must include proper exception handling for all invalid operations.
- Implement the logging decorator to log all key actions as demonstrated in class.
- Test your solution thoroughly against all the requirements listed below.
- Submit a single Python file (.py) with your complete solution.
- Do not use any external libraries — only Python's standard library is permitted.

■ Note

- This assignment assesses your ability to design a complete system using OOP, exception handling, and decorators — the core topics covered in this session.

Assignment: Library Management System

Design and implement a **Library Management System** in Python using Object-Oriented Programming. The system should allow librarians to perform the operations described below.

Q1. Add a Book

Implement functionality to add a new book to the library. Each book should store relevant details such as a unique Book ID, title, and author name.

Q2. Remove a Book

Implement functionality to remove an existing book from the library using its Book ID.

Q3. Search for a Book

Implement functionality to search for a book by its Book ID and display its details if found.

Q4. Display All Books

Implement functionality to display the details of all books currently available in the library.

Q5. Issue a Book to a Student

Implement functionality to issue an available book to a student. The system should record which student has been issued which book, along with the student's name and ID.

Q6. Return a Book

Implement functionality to process the return of a book that was previously issued to a student.

Q7. Display All Issued Books

Implement functionality to display a list of all books that are currently issued, including the details of the student to whom each book is issued.

Q8. Count Total Books

Implement functionality to count and display the total number of books currently in the library.

Q9. Prevent Duplicate Book IDs

Ensure that the system prevents two books from having the same Book ID. If a duplicate ID is detected, the system should handle it appropriately without crashing.

Q10. Handle Invalid Operations Using Exceptions

Implement proper exception handling for invalid operations. Examples include: attempting to remove a book that does not exist, attempting to issue a book that is already issued, attempting to return a book that was not issued, and any other invalid input or operation.

Q11. Log Important Actions Using a Decorator

Create a logging decorator and apply it to all key methods in your system (such as add, remove, issue, and return). The decorator should log when each action starts and when it finishes.